

Engineering Statistics

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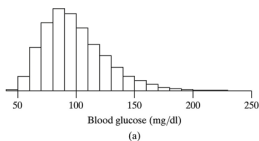
Chapter 4: Continuous Distribution

Outline

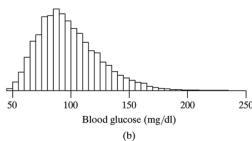
- ▶ Continuous random variables:
 - ▶ PDF, CDF
 - ▶ Expectation and Variance
- ▶ Common continuous Random variables:
 - ▶ Continuous uniform distribution
 - ▶ Exponential, Gamma and Weibull distribution
 - ▶ **Normal distribution**
 - ▶ Assessing normality
 - ▶ Normal approximation

Probability Density Curve

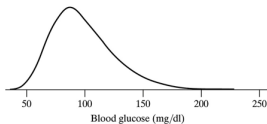
When the population corresponds to a **continuous** random variable, the probability distribution can be described by a **probability density function (PDF)** or a **probability density curve**.



(a)

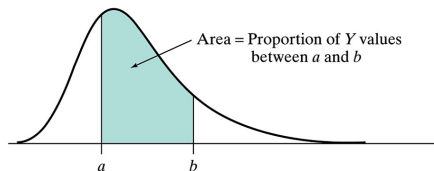


(b)



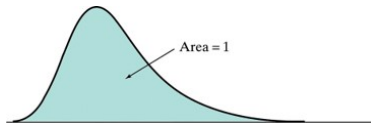
(c)

Probability Density Curve



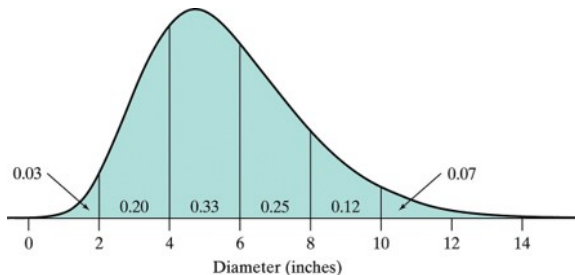
- For any two numbers a and b ,

Area under density curve between a and b
= Proportion of X values between a and b .



- The area under an entire density curve must equal 1.

Example: Diameters of 30-Year-Old Douglas Fir Trees



$$\begin{aligned} & \Pr(\text{Tree has diameter less than 4 **or** greater than 10 in}) \\ &= \Pr(0 \leq X \leq 2) + \Pr(2 \leq X \leq 4) + \Pr(10 \leq X \leq 12) \\ &= 0.03 + 0.2 + 0.07 = 0.3 \end{aligned}$$

Probability Density Functions

Let X be a continuous random variable, a **probability density function** (PDF), $f(x)$, is a function such that

1. $f(x) \geq 0$ for $x \in \mathbb{R}$.
 2. $\int_{-\infty}^{\infty} f(x)dx = 1$.
 3. $\Pr(a \leq X \leq b) = \int_a^b f(x)dx$ for any $a, b \in \mathbb{R}$.
- NOTE: It's clear that $\Pr(X = b) = \int_b^b f(x)dx = 0$.

Example

Consider the function

$$f(x) = \begin{cases} \frac{1}{6}x, & x \in [2, 4] \\ 0, & \text{elsewhere} \end{cases}.$$

► Q. Is it a probability density function?

1. $f(x) \geq 0$.
2. $\int_{-\infty}^{\infty} f(x)dx = \int_2^4 \frac{x}{6}dx = 1$.

Example

- ▶ Q. What is $\Pr(2 \leq X \leq 3)$?

$$\int_2^3 \frac{x}{6} dx = \frac{5}{12}.$$

- ▶ Q. What is $\Pr(X = 3)$?

X is a continuous random variable, so

$$\Pr(X = 3) = 0.$$

- ▶ Q. What is $\Pr(2 \leq X < 3)$?

Remark

- ▶ For **continuous** random variables,
 - ▶ $\Pr(X \leq x) = \Pr(X < x)$
 - ▶ $\Pr(X \geq x) = \Pr(X > x)$
 - ▶ $\Pr(X \leq x) = 1 - \Pr(X > x)$

Using AI to Learn: Probability from a PDF

Ask AI

Suppose $f(x) = x/6$ for $2 \leq x \leq 4$. Explain visually why

$$\Pr(2 \leq X \leq 3) = \frac{5}{12}.$$

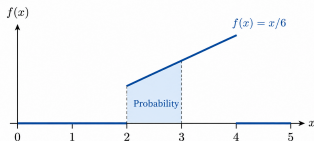
What does the integral represent?

AI Response

For a continuous random variable, probability is the **area under the PDF**.

$$\Pr(2 \leq X \leq 3) = \int_2^3 \frac{x}{6} dx = \frac{5}{12}.$$

So, the shaded area is $5/12$.



Integral = Area = Probability

The Cumulative Distribution Function

The **cumulative distribution function** (CDF) of a random variable X , denoted as $F(x)$, is

$$F(x) = \Pr(X \leq x), \quad \text{for any } x$$

► For **continuous** random variables X , $F(x)$ satisfies the following properties:

1. $F(x) = \Pr(X \leq x) = \int_{-\infty}^x f(t)dt$ for $x \in \mathbb{R}$.
2. $0 \leq F(x) \leq 1$.
3. If $x \leq y$, $F(x) \leq F(y)$.

Example

In our earlier example,

$$f(x) = \frac{1}{6}x, \quad x \in [2, 4].$$

Then,

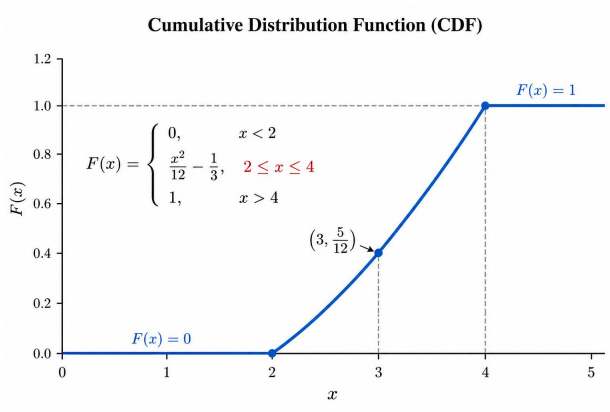
$$F(x) = \int_{-\infty}^x f(t)dt = \int_2^x \frac{t}{6}dt = \frac{x^2}{12} - \frac{1}{3}.$$

The cumulative distribution function is therefore:

$$F(x) = \begin{cases} 0, & x < 2 \\ \frac{x^2}{12} - \frac{1}{3}, & x \in [2, 4] \\ 1, & x > 4 \end{cases}$$

Example

$$F(x) = \frac{x^2}{12} - \frac{1}{3}, \quad x \in [2, 4]$$



Example

$$F(x) = \frac{x^2}{12} - \frac{1}{3}, \quad x \in [2, 4]$$

1. $\Pr(X \leq 3) = F(3) = \frac{5}{12}$.
2. $\Pr(X > 3) = 1 - F(3) = \frac{7}{12}$.
3. $\Pr(1 \leq X < 3) = F(3) - F(1) = F(3) - 0$.

Remark

For the continuous case, with the given CDF, $F(x)$, we can derive the PDF, $f(x)$:

$$f(x) = \frac{d}{dx} F(x).$$

Example:

Consider the following CDF

$$F(x) = \begin{cases} 0, & x \leq 0 \\ 1 - e^{-x/2}, & x > 0. \end{cases}$$

Then,

$$f(x) = \frac{d}{dx} F(x) = \frac{1}{2} e^{-x/2}, \quad x > 0.$$

Mean and Variance of a Continuous RV

Let X be a continuous random variable with density $f(x)$, and $g(X)$ be a function of the continuous random variable X . Then, the expected value of $g(X)$ is given by

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x)dx$$

provided that $\int_{-\infty}^{\infty} |g(x)|f(x)dx$ is finite.

Mean and Variance of a Continuous RV

Population mean: when $g(X) = X$,

$$\mu = E(X) = \int_{-\infty}^{\infty} xf(x)dx.$$

Population variance: when $g(X) = (X - \mu)^2$,

$$\sigma^2 = \text{Var}(X) = E[(X - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 f(x)dx.$$

Example

Consider the continuous random variable X with density

$$f(x) = \begin{cases} 0, & x \leq 0 \\ 7e^{-7x}, & x > 0. \end{cases}$$

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} xf(x)dx = \int_0^{\infty} x \cdot 7e^{-7x} dx \\ &= -xe^{-7x} \Big|_0^{\infty} - \int_0^{\infty} -e^{-7x} dx \quad (\text{integration by parts}) \\ &= -xe^{-7x} - \frac{1}{7}e^{-7x} \Big|_0^{\infty} \\ &= (0 - 0) - (0 - \frac{1}{7}) \\ &= \frac{1}{7}. \end{aligned}$$

Properties

Let X be a random variable (**discrete** or **continuous**) and let a, b be any constants.

1. $E[a] = a$.
2. $E[aX + b] = aE[X] + b$.
3. $\text{Var}[a] = 0$.
4. $\text{Var}[aX + b] = a^2 \text{Var}[X]$.

► **Note:** All of the properties of expectation apply to both discrete and continuous random variables.

Percentiles & Quantiles

By now you are probably well aware of what it means to be in the **90th percentile** on a test such as the SAT. This means that the score that you received is greater than or equal to 90% of all scores that were earned on that test.

Percentiles & Quantiles

- ▶ The **q -th percentile** is the value that is greater than $q \times 100\%$ of the values and less than $(1 - q) \times 100\%$ of the values in the population distribution.
- ▶ A **quantile** is simply a percentile represented as a decimal instead of a percentage (e.g. 50th percentile = 0.5 quantile).
- ▶ We can find quantiles using the CDF

Example

- Q. What value of x is greater than 50% of all values in the population? That is, find q such that

$$\Pr(X \leq q) = 0.5.$$

For the random variable X with PDF $f(x) = x/6$ for $x \in [2, 4]$, the CDF is

$$F(x) = \Pr(X \leq x) = \frac{x^2}{12} - \frac{1}{3}.$$

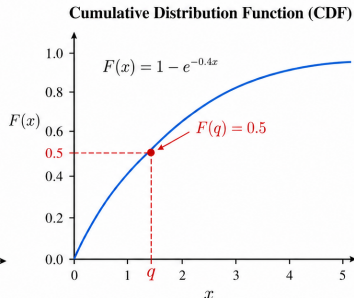
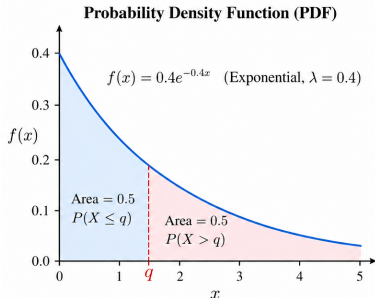
We want to find q such that $F(q) = 0.5$, so

$$\frac{q^2}{12} - \frac{1}{3} = 0.5 \Rightarrow q \approx 3.16.$$

Remark

- The quantile q such that $F(q) = 0.5$ is the median.

The quantile q such that $F(q) = 0.5$ is the median.



Median

The value q such that $F(q) = 0.5$ is the median. It divides the distribution into two equal halves:

$$P(X \leq q) = 0.5 \quad \text{and} \quad P(X > q) = 0.5.$$

Continuous Uniform distribution

Continuous Uniform distribution:

X is a uniform random variable on the interval (a, b) , denoted $X \sim U(a, b)$, if it has the PDF

$$f(x) = \begin{cases} \frac{1}{b-a}, & x \in (a, b) \\ 0, & \text{elsewhere.} \end{cases}$$

► Mean and variance:

$$E(X) = \frac{a+b}{2}, \quad \text{Var}(X) = \frac{(b-a)^2}{12}.$$

Exponential distribution

The exponential distribution is often used to model the time that elapses before an event occurs. It has a close connection the Poisson distribution.

The PDF of an exponential distribution with parameter $\lambda > 0$ (λ is called the rate parameter) is

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x > 0 \\ 0, & x \leq 0. \end{cases}$$

Exponential distribution

Notation:

$$X \sim \text{Exp}(\lambda)$$

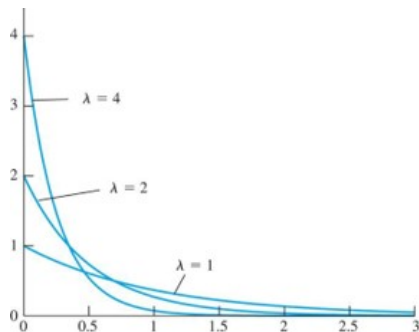
► CDF:

$$F(x) = \int_0^x f(t)dt = 1 - e^{-\lambda x}, \quad x > 0$$

► Mean and variance:

$$E(X) = \frac{1}{\lambda}, \quad \text{Var}(X) = \frac{1}{\lambda^2}.$$

Exponential distribution



The shape of the exponential distribution is determined entirely by λ .

Exponential and Poisson distribution

- ▶ Let X be the number of independent events in a t unit of time or space, then X follows a Poisson distribution with rate parameter λt .

$$X \sim \text{Pois}(\lambda t)$$

- ▶ Let T represents the waiting time from any starting point to the next event.
 - ▶ Let the starting point to be $t = 0$

Exponential and Poisson distribution

$$\begin{aligned}\Pr(T > t) &= \Pr(\text{Waiting at least } t \text{ time for the event to happen}) \\ &= \Pr(\text{No event in } t \text{ time}) \\ &= \Pr(X = 0 | T = t) \\ &= \frac{e^{-\lambda t} (\lambda t)^0}{0!} \\ &= e^{-\lambda t}.\end{aligned}$$

$$\therefore T \sim \text{Exp}(\lambda).$$

- $E(T) = \frac{1}{\lambda}$ is the mean amount of time/space until the next event.

Example

Suppose in a manufacturing facility, on average an emergency shutdown occurs once every 20 days. In observing this facility, what is the probability that we will have to wait as most 15 days until the next emergency shutdown?

Example

Let X be the number of shutdown in 20 days, then

$$X \sim \text{Pois}(\lambda t) = \text{Pois}(1).$$

Let T be the the time at which the first emergency shutdown occurs. Then

$$T \sim \text{Exp}(1/20).$$

The CDF for T is

$$F(t) = 1 - e^{-t/20}.$$

We wish to know

$$\Pr(T \leq 15) = F(15) = 1 - e^{-15/20} = 0.5276.$$

Memoryless property

If $T \sim \text{Exp}(\lambda)$ and $s, t > 0$, then

$$\Pr(T > t + s | T > s) = \Pr(T > t).$$

Memoryless property:

- ▶ The distribution **does not remember** how much time has elapsed since the previous event.
- ▶ If s units of time already have passed, then the probability that the next event will occur in an additional t units of time is the same as the probability that the event will occur in the next t units of time.

Example

Let's say the average waiting time for a bus is 10 minutes ($\lambda = 1/10$). Suppose you've been waiting for a bus for 15 minutes, and it hasn't arrived yet. Now, you want to know the probability that the next bus will arrive within the next 5 minutes. The probability remains the same whether you've just arrived at the bus stop or have been waiting for 15 minutes. That is,

$$\Pr(T > 20 | T > 15) = \Pr(T > 5).$$

Gamma function

The Gamma function, denoted Γ , is defined by

$$\Gamma(r) = \int_0^{\infty} x^{r-1} e^{-x} dx, \quad r > 0.$$

Properties:

- ▶ For an integer r , $\Gamma(r) = (r-1)!$
- ▶ For any r , $\Gamma(r) = (r-1)\Gamma(r-1)$
- ▶ $\Gamma(1/2) = \sqrt{\pi}$

Gamma distribution

- ▶ The Gamma distribution is an extension of the exponential distribution to model the waiting time until r events occur.
- ▶ If r is any positive integer, then the waiting time until r events occurs is distributed as a gamma distribution with parameters r , and λ , denoted

$$X \sim \Gamma(r, \lambda)$$

- ▶ When r is a positive integer, $\Gamma(r, \lambda)$ is often called Erlang distribution.

Gamma distribution

- PDF:

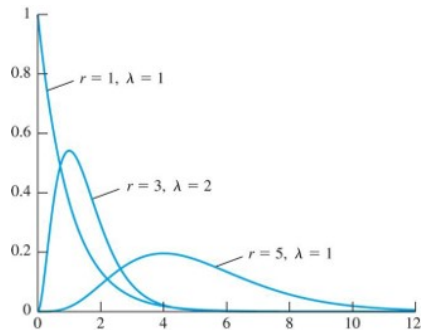
$$f(x) = \begin{cases} \frac{\lambda^r}{\Gamma(r)} x^{r-1} e^{-\lambda x}, & x > 0 \\ 0, & x \leq 0. \end{cases}$$

where $r, \lambda > 0$.

- Mean and variance:

$$E(X) = \frac{r}{\lambda}, \quad \text{Var}(X) = \frac{r}{\lambda^2}.$$

Gamma distribution



Special cases

1. $Exp(\lambda)$

- ▶ $\Gamma(1, \lambda) = Exp(\lambda)$.
- ▶ Suppose $Y_1, \dots, Y_r$ are distributed as $Exp(\lambda)$ and independent, then

$$X = Y_1 + \dots + Y_r \sim \Gamma(r, \lambda).$$

2. Chi-square distribution

- ▶ For $k \in \mathbb{N}$, $\Gamma(k/2, 1/2)$ is called Chi-square distribution with k degrees of freedom.
- ▶ Usually denoted $X \sim \chi_k^2$.

The Weibull Distribution

A random variable X follows a Weibull distribution with shape parameter $\beta > 0$ and scale parameter $\eta > 0$, written

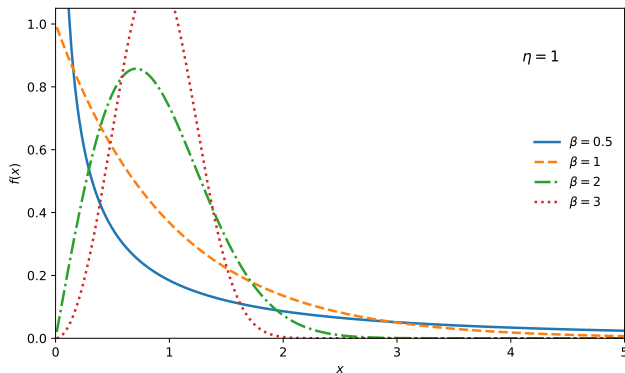
$$X \sim \text{Weibull}(\eta, \beta),$$

if its probability density function is

$$f(x) = \begin{cases} \frac{\beta}{\eta} \left(\frac{x}{\eta}\right)^{\beta-1} \exp \left[- \left(\frac{x}{\eta}\right)^{\beta} \right], & x \geq 0, \\ 0, & x < 0. \end{cases}$$

- ▶ β is the **shape parameter**.
- ▶ η is the **scale parameter**.
- ▶ Commonly used to model lifetimes and failure times.

Weibull PDF for Different Shape Parameters



- ▶ Changing β changes the **shape** of the distribution.
- ▶ When $\beta = 1$, the Weibull PDF becomes exponential.

Connection with the Exponential Distribution

The exponential distribution is a **special case** of the Weibull distribution.

If

$$X \sim \text{Weibull}(\eta, 1),$$

then

$$f(x) = \frac{1}{\eta} \exp\left(-\frac{x}{\eta}\right), \quad x \geq 0,$$

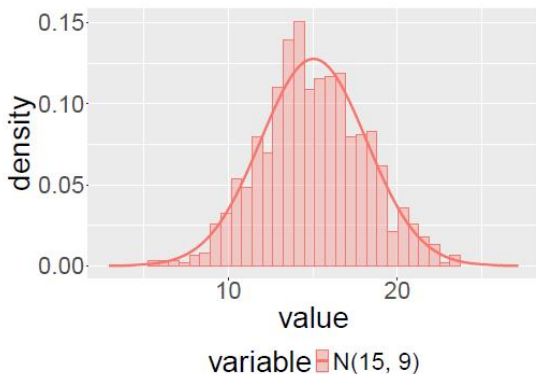
which is exponential with rate $\lambda = \frac{1}{\eta}$.

- ▶ $\beta < 1$: decreasing failure rate.
- ▶ $\beta = 1$: constant failure rate — **exponential**.
- ▶ $\beta > 1$: increasing failure rate.

Weibull extends exponential by allowing the failure rate to change over time.

Normal distribution

Normal (or Gaussian) distributions are a family of symmetric, unimodal and bell-shaped density curves.



If X follows a normal distribution with mean μ and standard deviation σ , then we denote $X \sim N(\mu, \sigma^2)$. - R uses an alternative notation: $X \sim N(\mu, \sigma)$.

Normal distributions

► PDF:

$$\begin{aligned}f(x) &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \\&= \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right).\end{aligned}$$

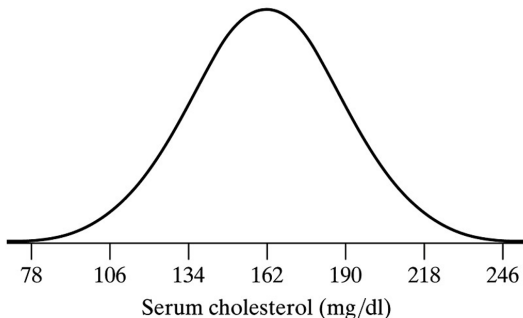
- $x \in (-\infty, \infty)$
- $\mu \in (-\infty, \infty)$
- $\sigma^2 > 0$.

► Mean and variance:

$$E(X) = \mu, \quad \text{Var}(X) = \sigma^2.$$

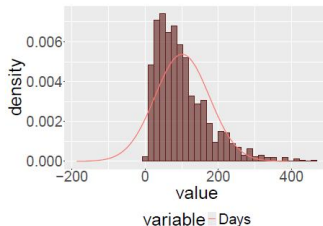
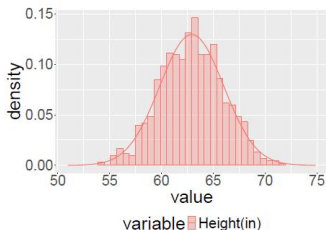
Example

Normal curves are used to model many continuous variables.
Let X be the serum cholesterol:



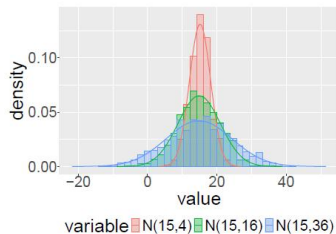
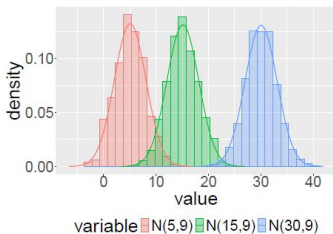
Normal distribution of serum
cholesterol, with $\mu = 162$ mg/dl and $\sigma = 28$ mg/dl

Example



- ▶ Human heights can be modeled by a Normal distribution.
- ▶ Guinea pigs survival times after inoculation of a pathogen would not be a good candidate for a Normal distribution model. Instead, it can be modeled by an Exponential distribution.

The Shape of the Normal Distribution



- ▶ Increase $\mu \rightarrow$ shift to the right
- ▶ Decrease $\mu \rightarrow$ shift to the left
- ▶ Larger $\sigma \rightarrow$ more spread out
- ▶ Smaller $\sigma \rightarrow$ more close together

Probabilities of $N(\mu, \sigma^2)$

To find probabilities, we need to find

$$\Pr(a \leq X \leq b) = \int_a^b f(x) dx.$$

However, there is no closed form solution for

$$\int_a^b \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx.$$

Someone has already done the approximation using the **standard Normal distribution**, so we can simply use either $\mathbb{R}$ or Z -Table to obtain the probabilities we want.

The Standard Normal Distribution

Let $X \sim N(\mu, \sigma^2)$. We can derive a new random variable Z by a linear transformation called standardization:

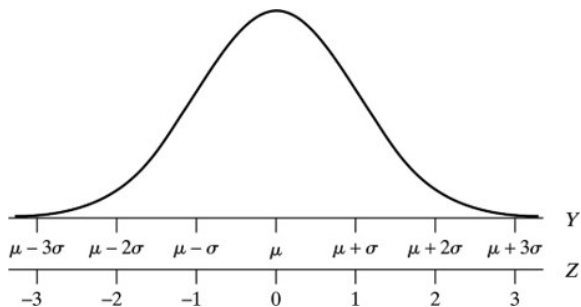
$$Z = \frac{X - \mu}{\sigma}.$$

Then, Z is also a normally distributed random variable with mean 0, and standard deviation 1, that is

$$Z \sim N(0, 1).$$

Z is called the **standard normal random variable**, and the corresponding distribution is called the **standard normal distribution**.

The Standard Normal Density Curve



z-scores

A **z-score** measures the number of standard deviations that the data value x is from μ .

$$Z = \frac{X - \mu}{\sigma} \implies X = \mu + \sigma Z.$$

Using z-score, we can find the the areas under the normal density curve. Let $X \sim N(\mu, \sigma^2)$, then

$\Pr(a < X < b) = \text{area under the curve between } a \text{ and } b.$

z-scores

We find the z-score for a and b . For the standard normal distribution $Z \sim N(0, 1)$, the z-scores for a and b are:

$$z_a = \frac{a - \mu}{\sigma}, \quad z_b = \frac{b - \mu}{\sigma}.$$

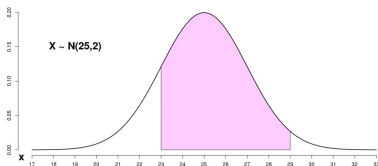
Then,

$$\begin{aligned} \Pr(a < X < b) &= \Pr(z_a < Z < z_b) \\ &= \text{area under the curve between } z_a \text{ and } z_b. \end{aligned}$$

Example

Let $X \sim N(25, 2^2)$.

$$\Pr(23 \leq X \leq 29) = \Pr(-1 \leq Z \leq 2).$$



Three Ways to Find Normal Probabilities

1. Use a z-table

Standardize $Z = (X - \mu)/\sigma$, then look up the probability in a standard normal table.

2. Use software like R

```
pnorm(x, mean = mu, sd = sigma)
```

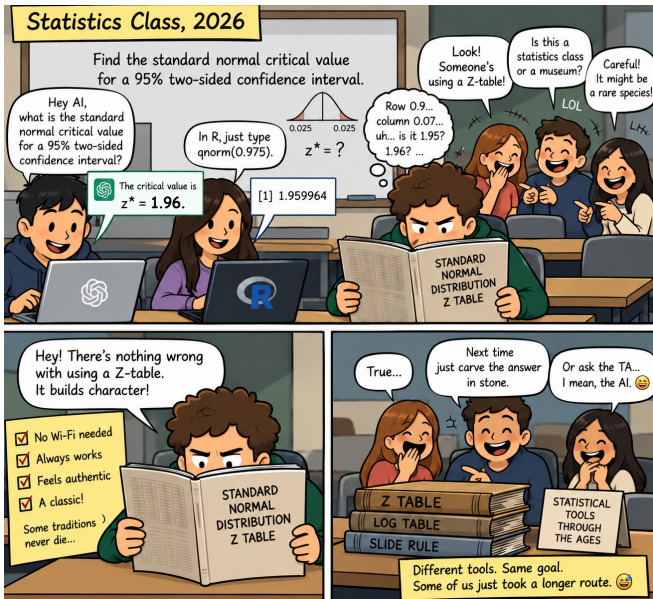
3. Ask AI

Ask AI

If $X \sim N(15, 9)$, what is $P(X \leq 18)$? Explain how you obtained the answer.

Different tools, same probability. Always understand and verify the result.

No Longer Teach the z-Table



Using R to obtain Probabilities

R provides a normal probability function called `pnorm()` that allows us to obtain probabilities.

```
pnorm(q, mean = 0, sd = 1, lower.tail = TRUE )
```

- ▶ If `mu` and `sigma` are not specified, then R assumes `mean = 0`, and `sd = 1`.
- ▶ We can set `lower.tail = FALSE` to get right-tail probabilities.

Example

Let $Y \sim N(54, 4.5^2)$. Find $\Pr(Y < 60)$.

1. Using z-table, $\Pr(Y < 60) = \Pr(Z < 1.33) = 0.90824$
2. Using R,

```
pnorm(60, mean = 54, sd = 4.5)  
pnorm((60-54)/4.5)
```


Example

The lifetime of a particular type of battery has a population distribution that follows a normal distribution with mean 50 hours and standard deviation 5 hours. Find the probability that a randomly chosen battery lasts between 42 and 52 hours.

Example

Let X = battery life in hours, then $X \sim N(50, 5^2)$.

$$\begin{aligned}\Pr(42 < X < 52) &= \Pr\left(\frac{42 - 50}{5} < \frac{X - 50}{5} < \frac{52 - 50}{5}\right) \\ &= \Pr(-1.6 < Z < 0.4) \\ &= \Pr(Z < 0.4) - \Pr(Z < -1.6) \\ &= 0.6006.\end{aligned}$$

Q. What is the probability that a randomly chosen battery lasts exactly 60 hours?

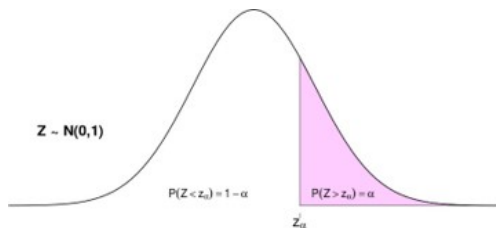
Quantiles of $N(\mu, \sigma^2)$

Assume that the battery lifetime is Normal distributed with mean 50 hours and standard deviation 5 hours. How long are the 75% longest lifetime in this population?
i.e. Find x such that

$$\Pr(X > x) = 0.75.$$

We know μ , σ , and the area under the curve. We want x .

Quantiles of $N(\mu, \sigma^2)$



Some notation that we'll use often:

- ▶ The area to the right of z_α is α .
- ▶ e.g. $z_{0.05}$ cuts off 5% of the area in the right tail.

Quantiles of $N(\mu, \sigma^2)$

z_α is the $1 - \alpha$ quantile of Z , so

$$\Pr(Z \leq z_\alpha) = 1 - \alpha, \quad \Pr(Z > z_\alpha) = \alpha.$$

How to find the $1 - \alpha$ quantile of X ?

$$Z = \frac{X - \mu}{\sigma} \Rightarrow X = \mu + \sigma Z.$$

Example: Battery lifetime

The battery lifetime $X \sim N(50, 5^2)$. From Z -table,

$$\Pr(Z > z) = 0.75 \Rightarrow \Pr(Z \leq z) = 0.25 \Rightarrow z \approx -0.67$$

Then,

$$x = \mu + \sigma z = 50 + 5 \times (-0.67) = 46.65.$$

The 75% longest battery lifetime in this population is about 46.65 days or longer.

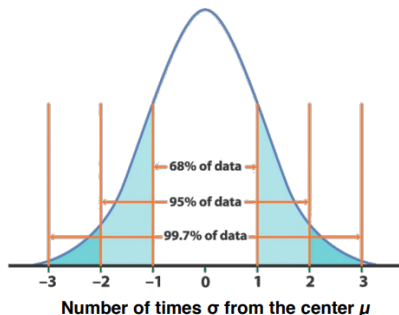
Using R,

```
qnorm(0.25, mean = 50, sd = 5)  
qnorm(0.25)*5 + 50
```

Some Normal Probabilities

For $X \sim N(\mu, \sigma^2)$:

- ▶ $\Pr(\mu - \sigma < X < \mu + \sigma) \approx 0.682$
- ▶ $\Pr(\mu - 2\sigma < X < \mu + 2\sigma) \approx 0.954$
- ▶ $\Pr(\mu - 3\sigma < X < \mu + 3\sigma) \approx 0.997$



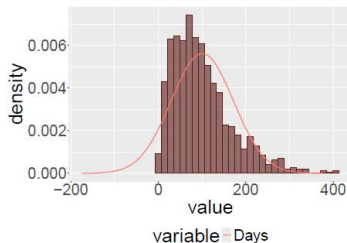
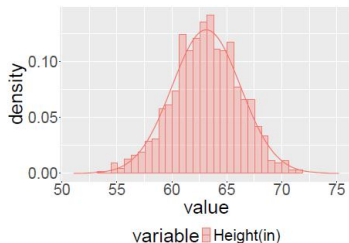
Assessing Normality

Suppose we have a sample, how do we know if that sample comes from a normal population or not?

1. Histogram
2. Normal probability plot (QQ plot)
3. Shapiro-Wilk Normality Test

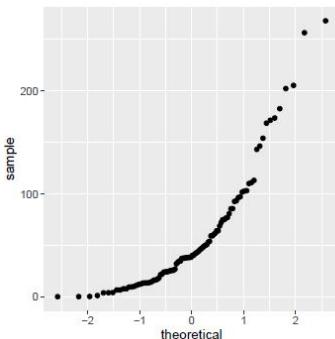
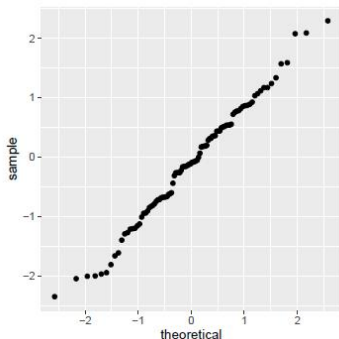
Assessing Normality: Histogram

- ▶ As the sample size becomes large, the histogram for the sample resembles the actual distribution of the population from which the sample was drawn.
- ▶ If this histogram is not bell-shaped, then the sample is most likely not from a normal distribution.



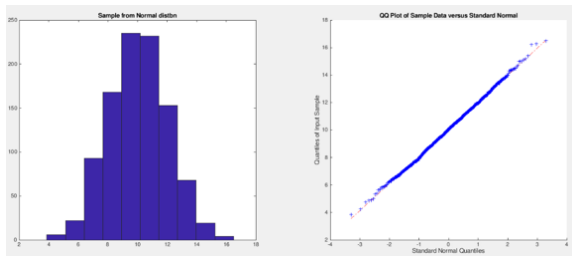
Assessing Normality: Normal Probability Plots

Another way to assess if a data set has an approximate Normal distribution is to plot the data on a normal probability plot (also called a normal **QQ plot** - stands for quantile-quantile).



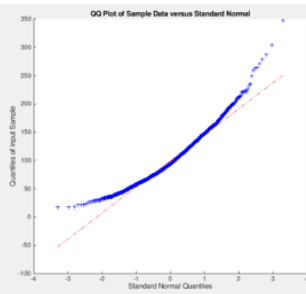
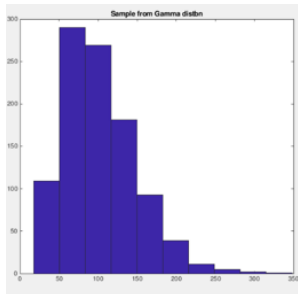
The way that these plots work is by comparing the observed quantiles, with the theoretical quantiles of a normal distribution.

Assessing Normality: Normal Probability Plots

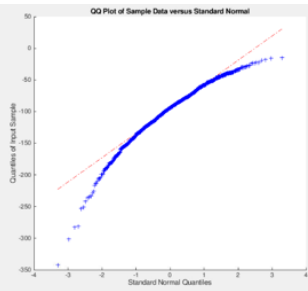
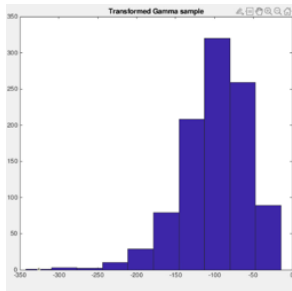


If a QQ plot produces a **straight line**, then the sample is **plausibly sampled from a normal distribution**.

Example: Skewed to the right

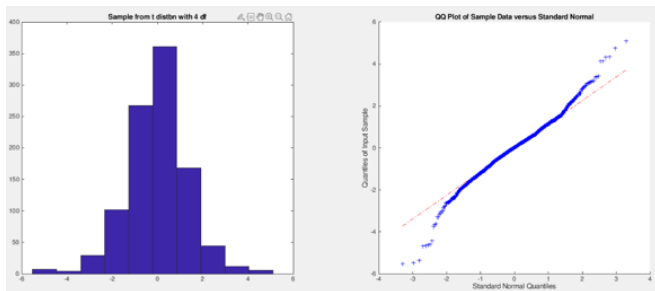


Example: Skewed to the left



Example: t -distribution

t -distribution is a probability distribution that resembles a normal distribution, but it has heavy tails.



Assessing Normality: Shapiro-Wilk Normality Test

- ▶ The Shapiro-Wilk test is a numerical test for the assessment of the normality of the data.
- ▶ The test outputs a quantity called a **p-value**.
 - ▶ If $p\text{-value} \leq 0.10$ then we have evidence that the data is not from a normal distribution.
 - ▶ If $p\text{-value} > 0.10$, we assume that the data is from a normal distribution.
- ▶ In R, `shapiro.test()` gives a test result.

Chapter Summary

- ▶ **Use PDFs and CDFs to describe continuous random variables.** Probabilities are areas under the PDF, while the CDF gives cumulative probabilities.
- ▶ **Use distribution summaries to describe location and variability.** Mean and variance describe center and spread, while quantiles identify values associated with specified cumulative probabilities.
- ▶ **Choose a distribution based on how the random variable arises.** Uniform models equally likely values; exponential and gamma model waiting times; Weibull models lifetimes; and normal models symmetric, bell-shaped data.
- ▶ **Use computing and AI to calculate probabilities and assess models.** R and AI can find probabilities and quantiles, while histograms, QQ plots, and normality tests help check whether a normal model is appropriate.